NEWTON SCHOOL OF TECHNOLOGY

WildChat Analytics Platform

Behavioral Intelligence & Safety Monitoring for Conversational AI at Scale

Team: Group G-6 (Group G-6)
Submission: April 2026 | Sector: AI / Tech

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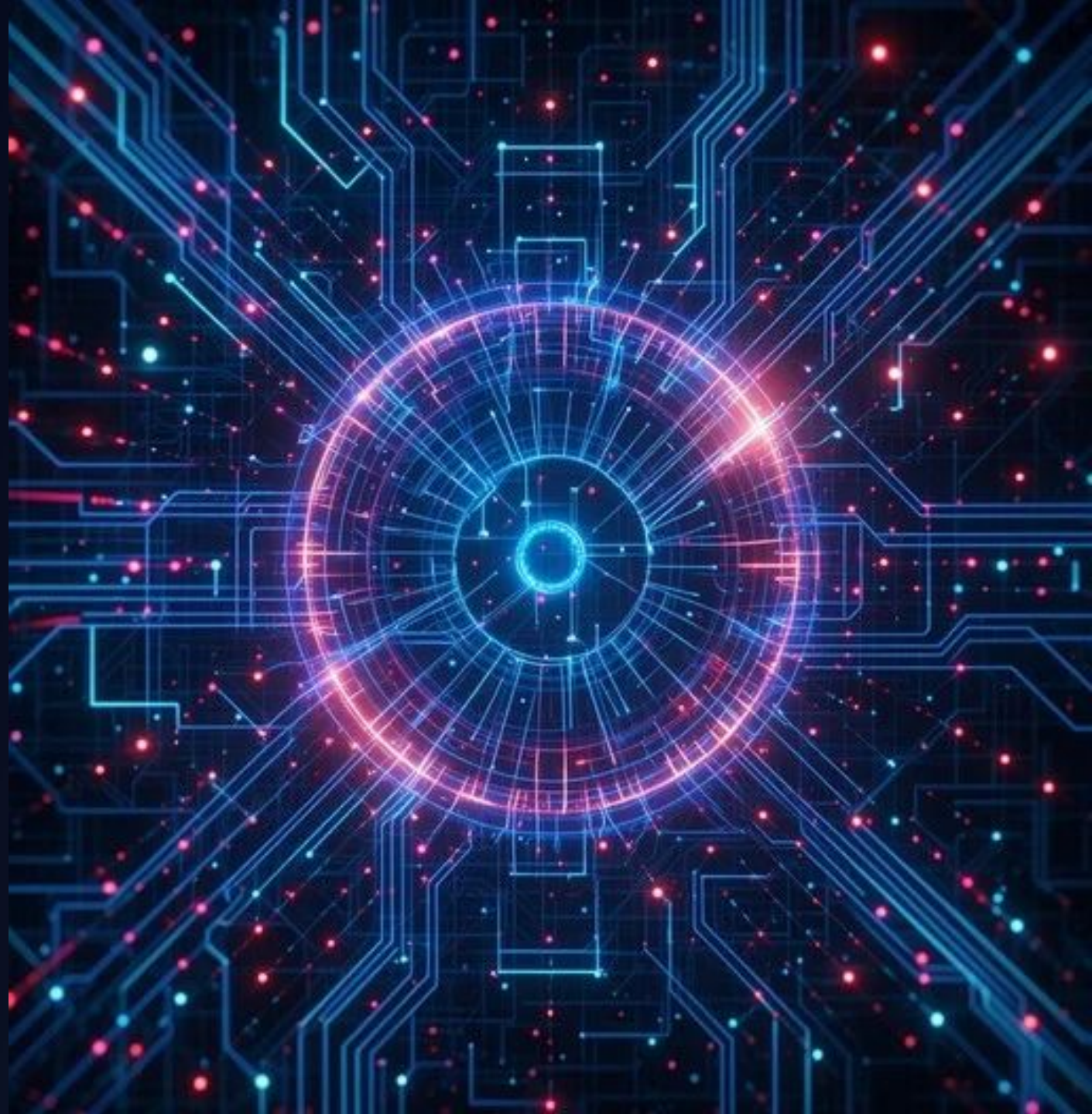


Context & Problem Statement

The AI Landscape

The global conversational AI market has surpassed **$10B+**, growing at a rapid **23% CAGR**. As adoption scales, operational blind spots emerge.

Problem Statement

Currently, there is no unified framework to monitor engagement, safety, and quality across multi-model LLM interactions.

Key Objectives

- Build a scalable ETL pipeline for conversational logs
- Develop behavioral analytics for engagement depth
- Launch Tableau dashboard for Safety Intelligence

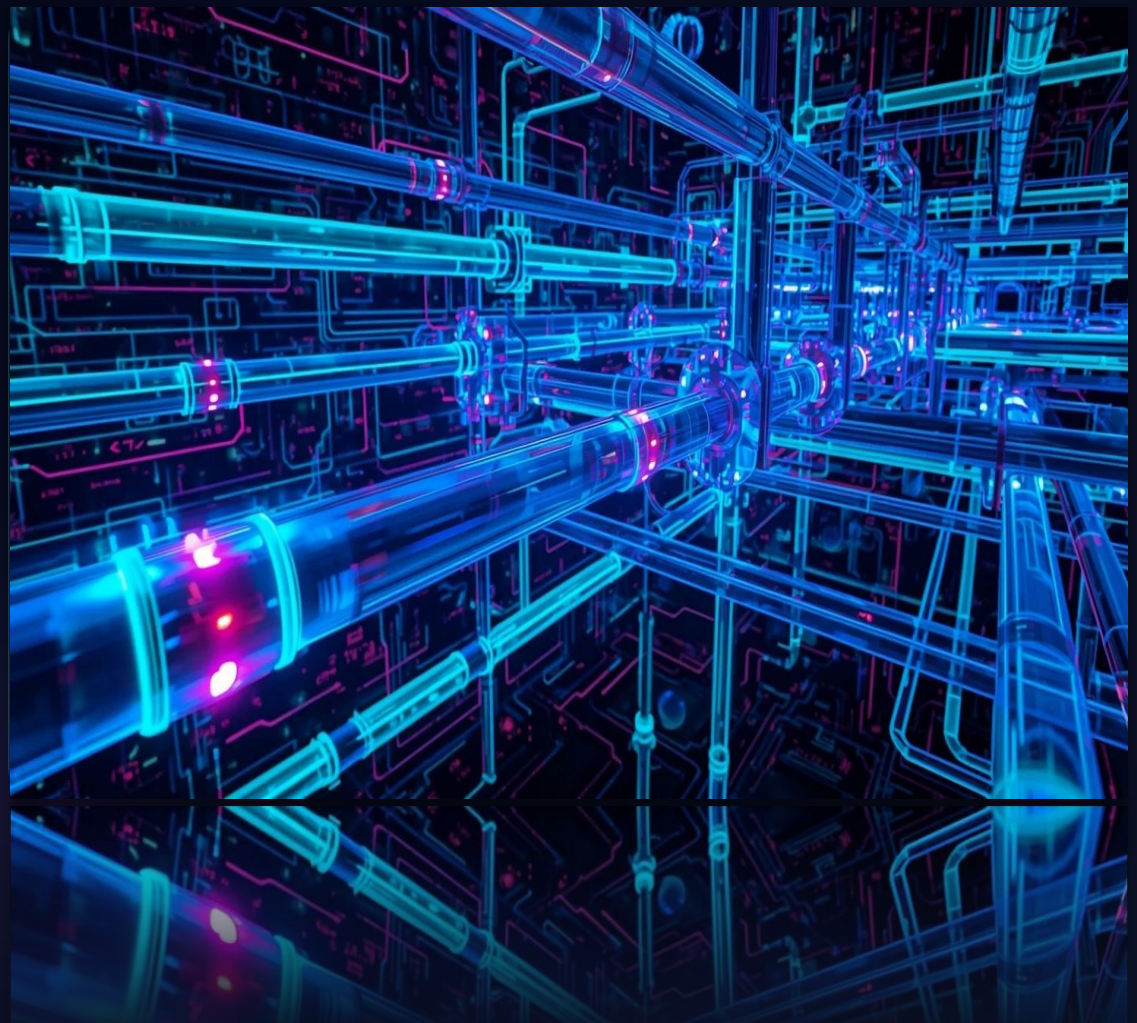
Stakeholders

PMs: Product Strategy & Retention

Safety: Policy Enforcement & Compliance

Researchers: Behavioral Data Insights

Data Pipeline Overview



Dataset Scale

Derived from **WildChat-1M** dataset, capturing real-world interactions across **68 languages** and **148 countries**.

49,550	6 Stages
Processed Records	ETL Pipeline

Processing Stack

Python / Pandas

NLTK / Scikit-learn

Steps: Ingestion → Cleaning → NLP → Feature Engineering → ML → Export

Platform KPI Framework

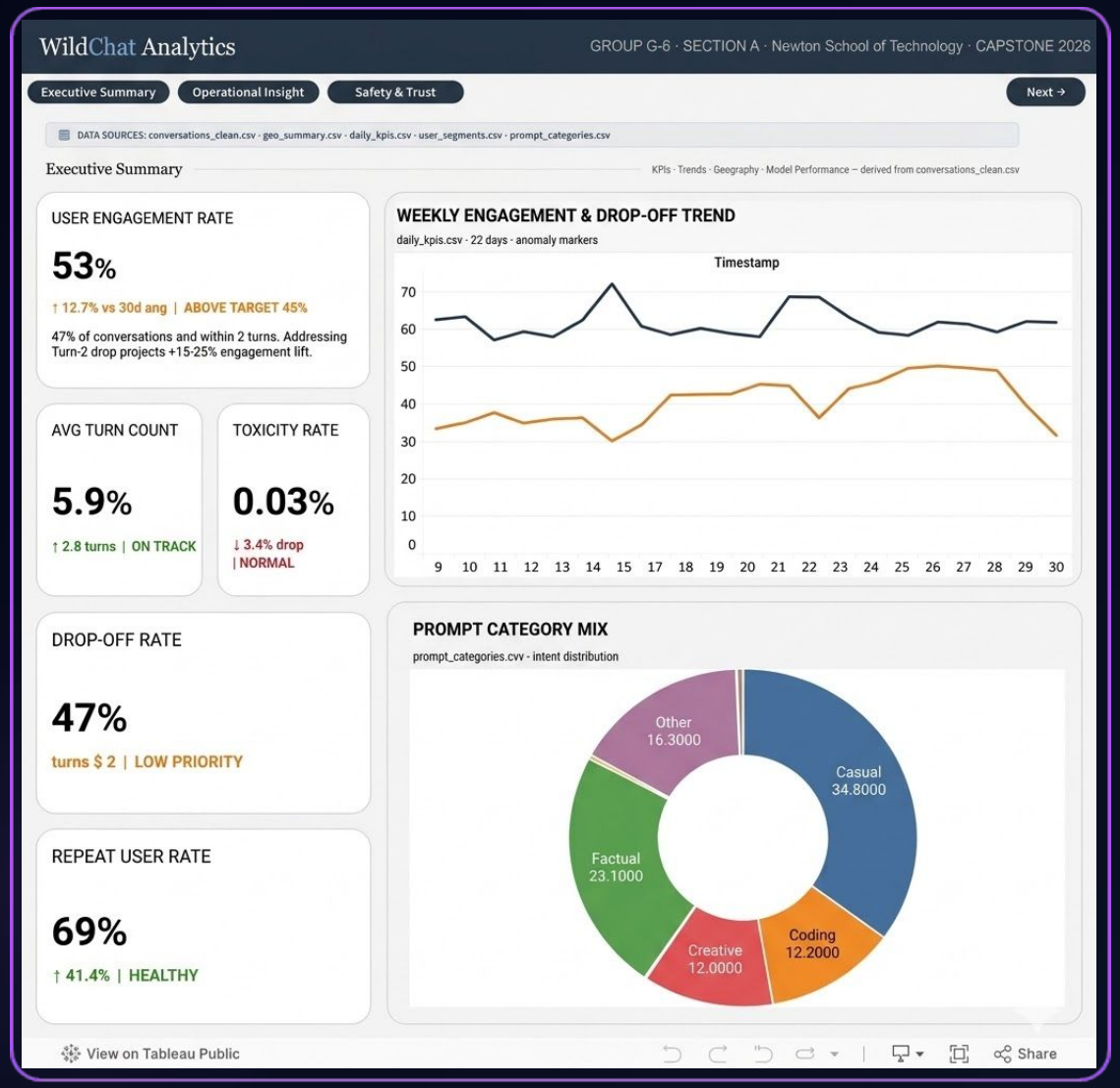
53% ENGAGEMENT RATE	**5.87%** AVG TURN COUNT	**0.03%** TOXICITY RATE	**47.2%** DROP-OFF RATE
69% REPEAT USER RATE	**+35%** LONG SESSION UPLIFT	**0.5x** LANGUAGE GAP	

Power Users (8% cohort) drive **59%** of total engagement turns.

Critical Platform Insights

- **Funnel Friction:** 47% of sessions drop at Turn 2; retention is a first-touch issue.
- **Pareto Concentration:** Just 8% of users drive 59% of all turns.
- **Regional Risk:** Germany toxicity is 5x the average; geo-targeted safety is needed.
- **Model Paradox:** GPT-4 yields 32% higher quality but 64% more toxic prompts.
- **Inclusion Gap:** Non-English users face significantly higher unanswered rates.
- **Sentiment Driver:** Creative prompts drive highest positive user satisfaction scores.

Tableau Intelligence Modules



Three Specialized Modules

Executive Summary

High-level KPIs, weekly trends, and platform health radar.

Operational Intelligence

Engagement funnels, language equity, and quality distributions.

Safety & Trust

Toxicity heatmaps, Z-score anomalies, and risk concentration.

Interactive Features: Global filters, scatter plot analysis, and heatmap drilling.

Strategic Recommendations

R1: Turn-2 Nudges

CRITICAL: Implement context-aware nudges after first response to bridge the Turn-2 drop-off gap.

R2: Geo-Safety

HIGH: Activate localized moderation thresholds for outliers like Germany and Brazil.

R3: Multi-Lingual

HIGH: Invest in quality parity for Chinese and Russian user segments (35% share).

R4: Loyalty UI

MEDIUM: Build power-user features to solidify the 8% core driving platform volume.

R5: Model Gating

MEDIUM: Calibrate separate safety layers for GPT-4 to handle adversarial probing.

R6: Drop off Reduction

MEDIUM: Targeting 15 - 20 % Reduction in Drop offs.

Estimated Business Impact

15-25%
RETENTION LIFT

30%
MODERATION SAVINGS

20%
CHURN REDUCTION

20-30%
SAFETY IMPROVEMENT

FOCUS AREA	BASELINE	TARGET OUTCOME	STRATEGIC VALUE
Turn-2 Retention	47.24% Drop-off	35.0% Drop-off	~10k New Sessions / Day
Global Moderation	Reactive Static Policy	Proactive Risk-Based	30% Reduction in Overhead
Multilingual Parity	57% Quality Gap	30% Quality Gap	Unlocks 17k+ Global Sessions
GPT-4 Safety	~37% False Negatives	~20% False Negatives	High Compliance Rating

Analytical Limitations

Data Constraints

- **Temporal Window:** 22-day snapshot limits seasonal trend analysis and seasonality logic.
- **Sample Size:** Message-level analysis constrained to ~500 sample records.
- **Geo-Bias:** IP geolocation prone to VPN distortion (especially in China).

Methodological Assumptions

- **Drop-off Proxy:** Turn-2 exit assumed as failure; some may be successful quick-answers.
- **Derived Quality:** Quality scores are model-calculated, not human-annotated.
- **Causation:** Observed correlations do not guarantee causal performance drivers.

Warning

The dataset window ends May 1, 2024. Market shifts in 2025/2026 behavioral patterns may differ.

Roadmap & Future Scope

Predictive ML

Develop gradient-boosted trees to predict drop-offs based on first-prompt semantic features.

Advanced NLP

Integrate BERTopic for deep semantic categorisation beyond current 5-cluster baseline.

Real-Time ETL

Transition from batch CSV processing to Kafka + Spark Streaming for live monitoring.

A/B Framework

Deploy automated experimentation to test retention nudges and localized safety policies.

Scaling architecture to handle 1M+ active daily conversation turns.

Team Group G-6 Contributions

MEMBER	PRIMARY RESPONSIBILITY	ARTIFACTS
Shaik Tajuddin	Project Lead & Dashboards	Tableau Dev / Report
Gaurav Meena	Visualization Design	UI/UX / Charts
Hemamdhar Nath	Data Acquisition	HuggingFace Hub Ingestion
Sushant Guri	Quality Engineering	Quality Score Logic
Jivit Rana	ETL Pipeline Architecture	Python Pipeline Scripts
Parthraj S. Bhati	Statistical Analysis	Anomaly / Z-Score Dev
Samarth Chaudhary	Communication & QA	PPT / Validation

Github

*Contributions verifiable via commit logs